

RANKIN INLET, NU, Canada

WMO: 710830

Lat: 62.810N

Lon: 92.110W

Elev: 106

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.8	(c) -39.1	(d) -47.9	(e) 0.3	(f) -41.6	(g) -45.8	(h) 0.4	(i) -39.0	(j) 42.0	(k) -15.3	(l) 38.7	(m) -19.4	(n) 12.9	(o) 350	(p) 1.092

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	67.9	57.2	63.5	54.6	59.8	52.6	57.9	66.2	55.4	62.4	53.1	58.9	12.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
53.9	62.0	60.8	51.3	56.3	58.2	49.4	52.4	55.5	25.1	66.5	23.5	62.6	22.1	59.1	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 36.1	(b) 31.4	(c) 28.2	(e) -43.7	(f) 75.4	(g) 3.8	(h) 3.8	(i) -46.5	(j) 78.2	(k) -48.7	(l) 80.4	(m) -50.9	(n) 82.6	(o) -53.6	(p) 85.4		
(a) 36.1	(b) 31.4	(c) 28.2	(e) -43.8	(f) 62.2	(g) 3.8	(h) 2.2	(i) -46.5	(j) 63.8	(k) -48.7	(l) 65.1	(m) -50.8	(n) 66.3	(o) -53.6	(p) 67.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	-21.7	-21.4	-11.8	4.7	22.9	40.5	52.0	50.2	40.0	25.2	3.4	-11.3		
	DBStd	28.29	11.97	11.31	13.30	12.07	9.14	6.20	5.85	4.68	6.10	8.75	13.87	13.45		
	HDD50	13102	2224	1999	1917	1359	841	292	44	55	304	770	1397	1900		
	HDD65	18403	2689	2419	2382	1809	1306	736	405	460	750	1235	1847	2365		
	CDD50	174	0	0	0	0	0	5	104	61	4	0	0	0		
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH74	15	0	0	0	0	0	1	13	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.5	15.0	14.9	14.7	14.5	14.3	12.8	12.4	13.3	14.8	16.3	15.7	15.1		
	PrecAvg	12.6	0.4	0.4	0.5	0.7	0.7	0.9	1.7	2.0	1.9	1.6	1.0	0.7		
(15) Precipitation	PrecMax	16.1	1.3	1.6	1.3	3.8	2.1	2.0	4.6	3.2	4.5	3.5	2.4	1.5		
	PrecMin	8.6	0.1	0.0	0.1	0.0	0.0	0.0	0.2	0.4	0.5	0.4	0.2	0.2		
	PrecStd	2.4	0.3	0.3	0.4	0.7	0.6	0.6	1.1	0.9	0.9	0.9	0.6	0.3		
	PrecStd	2.4	0.3	0.3	0.4	0.7	0.6	0.6	1.1	0.9	0.9	0.9	0.6	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	13.4	25.5	33.7	44.1	66.4	75.6	70.0	60.4	43.0	32.6	25.9		
		MCWB	12.8	13.1	25.2	32.6	39.1	55.3	59.9	58.7	52.8	41.7	32.0	24.8		
	2%	DB	7.5	5.8	18.2	29.4	38.7	58.5	70.7	65.0	53.7	39.8	29.9	19.1		
		MCWB	6.7	5.2	17.5	28.3	35.5	50.0	58.4	55.8	49.0	38.5	29.0	18.4		
	5%	DB	2.1	1.9	13.2	25.9	35.8	54.2	66.2	61.4	50.4	37.7	27.9	13.9		
		MCWB	1.2	1.2	12.4	24.8	33.5	47.1	55.9	53.9	46.8	36.5	27.1	13.2		
	10%	DB	-2.5	-2.5	8.5	22.9	33.9	50.2	62.1	58.0	47.9	35.5	25.4	9.3		
		MCWB	-3.3	-3.3	7.7	22.1	32.2	44.4	53.7	52.3	45.2	34.3	24.4	8.6		
	0.4%	WB	13.1	12.8	25.0	32.6	39.5	55.9	62.1	60.1	53.7	42.0	32.2	25.0		
		MCDB	13.6	13.1	25.3	33.4	43.3	65.6	71.7	67.0	58.5	42.8	32.3	25.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	6.7	5.2	17.5	28.5	35.8	50.4	59.1	57.1	49.8	38.8	29.3	18.5		
		MCDB	7.5	5.7	18.1	29.2	38.2	57.6	68.8	63.1	53.1	39.5	29.8	19.0		
	5%	WB	1.3	1.2	12.5	24.9	34.0	47.2	56.6	54.7	47.5	36.5	27.2	13.3		
		MCDB	2.0	1.8	13.3	25.7	35.5	53.6	65.2	60.0	49.6	37.4	27.9	13.9		
	10%	WB	-3.3	-3.3	7.8	22.0	32.4	44.7	54.2	52.6	45.5	34.5	24.5	8.6		
		MCDB	-2.5	-2.6	8.6	23.0	33.7	49.8	61.3	57.5	47.4	35.3	25.4	9.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.6	12.0	13.9	14.4	10.4	12.3	14.1	10.7	8.2	8.4	13.0	12.6		
		MCDBR	16.5	16.5	17.2	13.5	10.2	20.4	21.4	17.0	12.5	8.1	13.0	16.7		
		MCWBR	16.3	16.4	17.1	13.0	8.3	13.0	11.4	9.6	8.4	7.8	13.0	16.7		
		MCDBR	16.7	16.2	17.1	13.2	9.6	19.8	20.5	15.5	11.5	8.0	13.1	16.6		
	5% WB	MCWBR	16.4	16.1	17.1	12.8	8.0	13.1	11.9	9.8	8.3	7.8	13.2	16.6		
		MCWBR	16.4	16.1	17.1	12.8	8.0	13.1	11.9	9.8	8.3	7.8	13.2	16.6		
(40) Clear-Sky Solar Irradiance	taub		0.173	0.232	0.266	0.336	0.358	0.332	0.358	0.341	0.310	0.274	0.185	0.146		
	taud		2.000	2.166	2.180	1.998	2.061	2.420	2.438	2.488	2.533	2.372	2.034	1.916		
	Ebn at Noon		167	234	267	264	269	283	271	266	255	216	154	108		
	Edh at Noon		10	20	29	44	45	32	31	27	21	16	9	5		
(44) All-Sky Solar Radiation	RadAvg		73	351	898	1562	1870	1870	1761	1244	723	281	96	33		
	RadStd		9	22	62	89	141	138	127	95	61	33	15	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page